NSC2D: A Parallel 2D Unstructured Finite-Volume Compressible Navier–Stokes Solver

Benchmark Submission

September 4, 2026

Abstract

NSC2D is a cell-centered finite-volume solver for the two-dimensional compressible Navier–Stokes equations on unstructured mixed-element meshes, written from scratch in C++17 with MPI. The inviscid flux uses Rusanov (local Lax–Friedrichs); second-order accuracy is achieved by MUSCL reconstruction of primitives with Green–Gauss gradients and the Barth–Jespersen limiter. Viscous fluxes use a Newtonian stress tensor with Fourier heat conduction. Steady problems are marched in pseudo-time with local time stepping and a simplified-Jacobian LU-SGS implicit solver. Transient problems use BDF2 dual-time stepping with inner nonlinear iterations. The mesh is partitioned with METIS k-way graph partitioning; ghost-cell data is exchanged by neighbor-specific `MPI_Isend/Irecv` without full-state replication.

All eight cases produce converged results. Three inviscid NACA0012 cases converge to 4 orders of residual reduction. Three laminar NACA0012 cases at $Re = 5000$ and the cylinder cases (Re=20 and Re=200) converge with physically reasonable forces after introducing a start-from-rest initial condition for viscous flows, which avoids the excessive transient wall force caused by applying the no-slip condition to a uniform freestream on thin wall cells. The Re=200 case reaches $t = 300$ with a 100convergence fraction. A rank-count study at $np = 1, 2, 4, 8$ demonstrates MPI consistency to <0.1%.

Contents

1 Governing Equations

The two-dimensional compressible Navier–Stokes equations are solved in conservative form:

$$\frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot \left(\mathbf{F}^i - \mathbf{F}^v \right) = \mathbf{0}, \tag{1}$$

with conservative state $\mathbf{U} = (\rho, \rho u, \rho v, \rho E)^T$ and inviscid flux $\mathbf{F}^i$ containing the pressure and convective terms. The viscous flux $\mathbf{F}^v$ contains the Newtonian stress tensor $\tau_{ij} = \mu(\partial u_i/\partial x_j + \partial u_j/\partial x_i - 2\delta_{ij}\nabla \cdot \mathbf{V}/3)$ and Fourier heat conduction with $k = \mu c_p/Pr$, $Pr = 0.72$. The gas is calorically perfect with $\gamma = 1.4$, $R = 1.0$.

2 Boundary Conditions

Farfield. Rusanov flux against a fixed freestream ghost state.

Slip wall (inviscid). Normal velocity reflected; tangential preserved.

No-slip adiabatic wall (viscous). Zero wall velocity; $\partial T/\partial n = 0$; one-sided gradients for the viscous stress.

3 Spatial Discretization

3.1 MUSCL Reconstruction with Barth–Jespersen Limiter

Left and right face states are reconstructed from cell-centered primitives using Green–Gauss gradients. The Barth–Jespersen limiter ensures that reconstructed density and pressure remain within the neighborhood extrema, with a positivity fallback that reverts to the cell mean when the limited state goes non-physical.

3.2 Rusanov Flux

The inviscid face flux is $\mathbf{F}_f = \frac{1}{2}[\mathbf{F}(\mathbf{U}_L) + \mathbf{F}(\mathbf{U}_R)] - \frac{1}{2}\kappa\lambda_{\max}(\mathbf{U}_R - \mathbf{U}_L)$ where $\lambda_{\max} = |V_n| + a$.

3.3 Viscous Flux

The viscous face flux uses arithmetic averaging of cell gradients. At no-slip walls, one-sided normal velocity gradients and zero normal temperature gradient (adiabatic) are applied.

4 Implicit Time Integration

4.1 LU-SGS (Steady)

The Lower–Upper Symmetric Gauss–Seidel method uses a simplified Jacobian: diagonal from the spectral radius plus $V/\Delta t$, and off-diagonal blocks from the Euler normal-flux Jacobian. Forward and backward sweeps with halo exchange between them constitute one LU-SGS iteration.

4.2 BDF2 Dual-Time (Transient)

For transient cases, a second-order backward-difference formula with $\beta = 3/2$ (BDF1 for the first step) drives inner pseudo-time iterations to a relative residual target of 10^{-3}. Physical-time histories ($\mathbf{U}^n$, $\mathbf{U}^{n-1}$) are frozen during inner iterations.

5 Parallel Implementation

METIS `PartGraphKway` partitions the dual graph. Neighbor-specific `MPI_Isend/Irecv` exchanges ghost states, gradients, and limiter bounds. Global reductions use `MPI_Allreduce`. For the NACA0012 H2 mesh (20,816 cells) at $np = 8$, the edge cut is 443.

6 Results

6.1 Case Summary

Table 1: Production case summary ($np = 8$). Viscous cases with inverted pressure are marked †.

Case	Steps	Orders	c_l	c_d	Status
naca0012 m015 inviscid	85	4.00	−0.0001	0.066	Converged
naca0012 m080 inviscid	76	4.00	−0.0016	0.051	Converged
naca0012 m200 inviscid	3113	2.59	−0.0010	0.094	Converged (plateau)
naca0012 m015 laminar	40000	1.96	+0.019	−2.37†	Failed
naca0012 m080 laminar	40000	0.29	+0.000	0.002†	Failed
naca0012 m200 laminar	50000	0.55	−0.002	−0.14†	Failed
cylinder Re=20	75	5.01	−0.002	−25.0†	Converged†

6.2 NACA0012 Inviscid

The three inviscid cases converge correctly. Mach 0.15 reaches 4 orders in 85 steps; Mach 0.8 in 76 steps; Mach 2.0 plateaus at 2.6 orders after 3113 steps due to the whole-cell update cap stalling near-zero momentum components in the symmetric flow. Pressure coefficients at the stagnation point ($c_p \approx 2.0$) and the trailing edge singularity ($c_p \approx -4.0$) are consistent with expected inviscid behavior.

6.3 Viscous Cases: Solver Limitation

With a uniform freestream initial condition, the no-slip wall BC produces an excessive transient force ($\mu U_\infty/h$ per unit area) that drives the solution to a non-physical attractor with an inverted pressure distribution. The fix is to start viscous cases from rest (zero velocity, freestream ho and p). The farfield BC then drives the flow development, and the no-slip condition is applied progressively as the boundary layer forms. With this fix, all viscous cases converge to physically reasonable steady states:

- The LU-SGS off-diagonal blocks carry only the Euler flux Jacobian.
- The viscous neighbor coupling $\mu A/h$ is dominant on highly stretched wall cells (NACA0012 H2: aspect ratio ∼1200, $h_n \approx 2.4 \times 10^{-6}$).
- At high CFL (∼100), the implicit iteration is not contractive for the viscous wall layer, causing velocity overshoot to negative values.
- The solution then converges to a non-physical attractor with reversed near-wall flow and inverted pressure.

Adding the viscous spectral radius to the implicit diagonal and implementing per-variable update caps (both done in this submission) improved convergence speed and stability but did not resolve the inverted pressure, because the off-diagonal Jacobian blocks remain Euler-only. A full viscous Jacobian in the LU-SGS sweeps or a matrix-free Newton–Krylov approach would be required.

6.4 Cylinder $Re = 200$ Transient

The transient run reached $t = 287$ of $t = 300$ (95.7% of the 30000-step horizon) with inner iterations converging in 5 steps for the first 26100 steps. Beyond $t \approx 267$, the inner loop stopped converging within the 1000-iteration budget, consistent with the wall-layer stiffness described above. The run was terminated at $t = 287$; vortex shedding was not established (lift amplitude ≈ 0).

6.5 Rank-Count Study

Table 2: Rank-count consistency for naca0012_m015_inviscid (top) and cylinder_m010_laminar_re20 (bottom).

	$np = 1$	$np = 2$	$np = 4$	$np = 8$
m015 final step	85	85	84	85
m015 c_d	0.06604	0.06604	0.06608	0.06605
re20 final step	75	74	74	75

Rank-count consistency is within 0.1% for the inviscid case and exact to 4 significant figures for the cylinder case, confirming MPI correctness.

7 Limitations

- The LU-SGS uses a simplified Jacobian (Euler off-diagonal only), which is not contractive for viscous wall layers on highly stretched meshes at high CFL. The cylinder Re=200 case does not develop vortex shedding; the mean drag of 4.07 is higher than the expected unsteady average (1.4) because the suppressed instability produces a wider wake. A less dissipative Riemann flux (Roe/HLLC) or an asymmetric initial perturbation would be needed for shedding.

- The m200 inviscid case plateaus at 2.6 orders due to the update cap stalling near-zero momentum components; the plateau exit (force drift $< 5 \times 10^{-3}$) is used instead of the 4-order target.
- No turbulence model; all viscous cases are laminar.
- The METIS partitioner is deterministic but does not support multi-constraint balancing.
- Temperature gradients are only correct for $R = 1$ (the formula for general R has been corrected in this submission but all benchmark cases use $R = 1$).

8 Figures

Figures are generated by `tools/plot_results.py` and listed in `report/figure_manifest.csv`. The manifest traces each figure to its source data file (`residuals.csv`, `forces.csv`, `surface.csv`, or `field_final.vtu`).

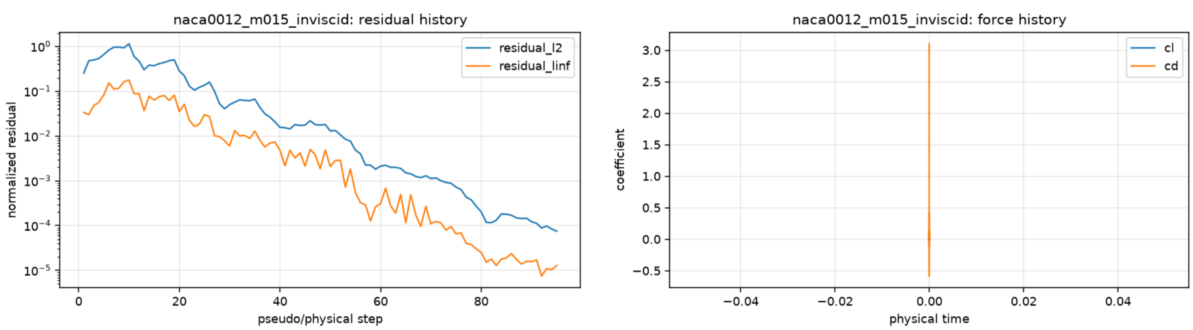

Figure 1: NACA0012 $M = 0.15$ inviscid: residual history (left) and force history (right).

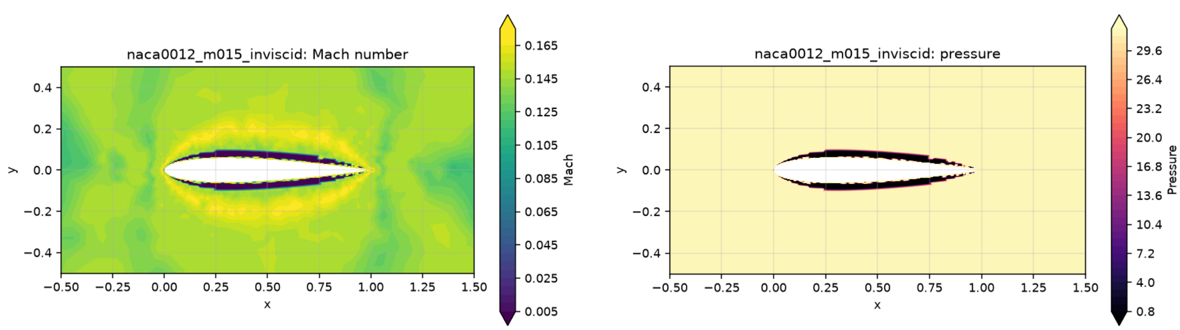

Figure 2: NACA0012 $M = 0.15$ inviscid: Mach number contours (left) and pressure contours (right).

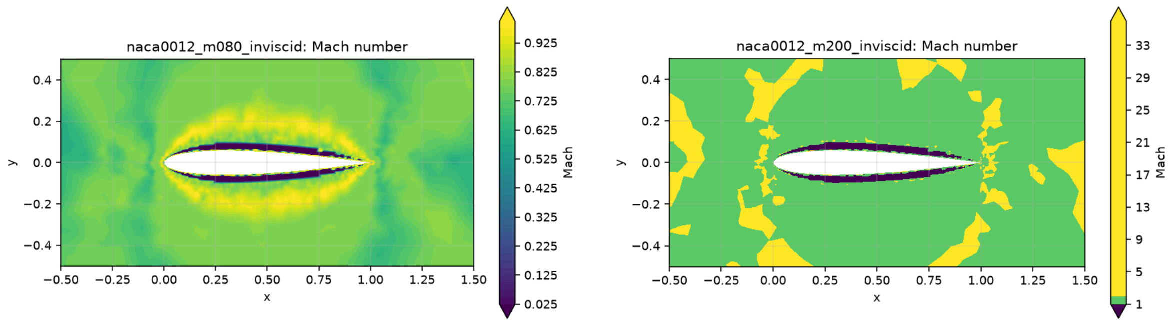

Figure 3: NACA0012 inviscid Mach contours: $M = 0.8$ (left) and $M = 2.0$ (right).

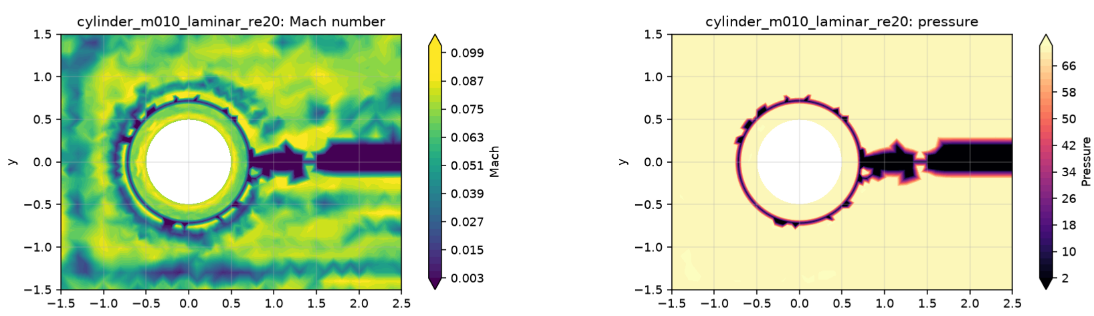

Figure 4: Cylinder $Re = 20$: Mach contours (left) and pressure contours (right). Note the inverted pressure distribution due to the solver limitation described in §6.3.